Multi-Qubit Systems

Quick Recap and Why We Need More Than One Qubit

A single qubit can exist in superposition, quantum gates manipulate qubits, quantum circuits describe quantum computations, and Qiskit provides the software framework for building and executing quantum programs. Today we move from one qubit to multiple qubits, unlocking exponentially larger state spaces.

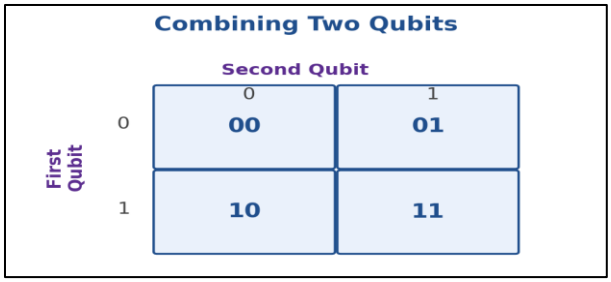
Imagine trying to write a paragraph using only one letter; no matter how many times you reuse it, you cannot express meaningful information. Similarly, a single qubit is useful for learning, but practical quantum algorithms such as quantum search, quantum cryptography, and error correction require many qubits working together. The real power of quantum computing begins when qubits interact.

1. From One Qubit to Two Qubits

A single qubit is written as |psi> = alpha|0> + beta|1>, where |alpha|^2 + |beta|^2 = 1. A common misconception is that two qubits simply give four states side by side. Not exactly: two qubits form one combined quantum system whose state is described using the tensor product, combining them into a single state space rather than two independent ones.

The Tensor Product (Intuition First)

Think of each qubit as adding a new dimension of information. When two qubits are combined, we do not add their possibilities, we multiply them. One qubit has 2 possible states, another qubit has 2 possible states, and together the possible basis states become 00, 01, 10, and 11: 2 x 2 = 4 possible basis states. This multiplication process is called the tensor product.



Mathematical Representation

The tensor product is represented using the symbol (x). For example, |0> (x) |1> = |01>, and |1> (x) |0> = |10>. In practice, the (x) symbol is usually omitted once the combined state is written.

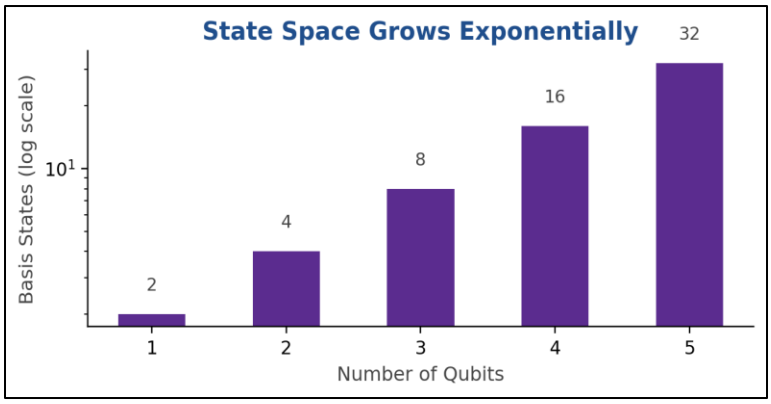
For two qubits, there are four computational basis states: |00>, |01>, |10>, |11>. Each basis state represents one definite outcome when both qubits are measured.

2. State Vector of Two Qubits

Unlike a single qubit, a two-qubit system requires four probability amplitudes. The general state is |psi> = alpha|00> + beta|01> + gamma|10> + delta|11>, where |alpha|^2 + |beta|^2 + |gamma|^2 + |delta|^2 = 1. This is simply the same normalisation rule extended to four basis states.

How Does the State Space Grow

One of the most remarkable properties of quantum computing is exponential growth. Every additional qubit doubles the size of the state space, which is one of the reasons quantum computers can outperform classical computers for certain problems.



3. Visualising Multi-Qubit States

Instead of a single Bloch Sphere, a multi-qubit system is imagined as several qubits combined through the tensor product, each contributing its own two basis states to the combined system.

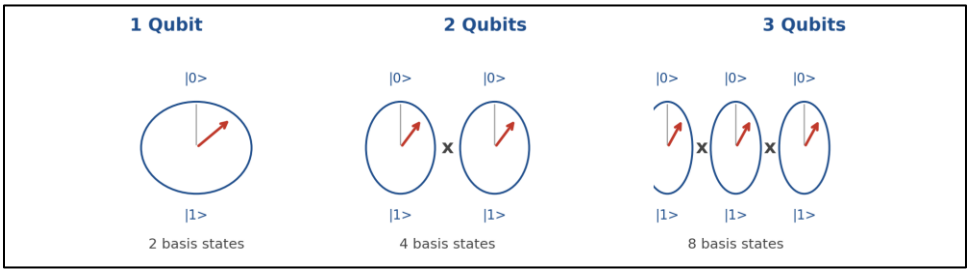


Figure 1: Combining n single-qubit systems multiplies their state spaces to 2^n basis states.

4. Multi-Qubit Circuits in Qiskit

Creating multiple qubits is straightforward, and you can apply different gates to each qubit independently.

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(2, 2)
qc.h(0)
qc.x(1)
```

Here, the first qubit enters superposition, while the second qubit undergoes a bit flip. This creates 2 qubits and 2 classical bits.

Measuring Multiple Qubits

Measurement is performed independently for each qubit, for example qc.measure([0,1],[0,1]). Possible outputs include 00, 01, 10, or 11, and the exact probabilities depend on the operations performed before measurement.

Worked Example

Suppose both qubits begin in |00>. Apply H on q0 and X on q1. After these operations, the first qubit is in superposition while the second is fixed at |1>. When measured, the system can produce |01> or |11>, each with approximately equal probability. This demonstrates how each qubit contributes independently to the overall system when no entangling operations are applied.

5. Why This Matters

With multiple qubits, quantum computers can represent exponentially larger state spaces, perform parallel quantum operations through superposition, prepare for entanglement and controlled operations, and execute practical quantum algorithms. This is the point where quantum computing begins to show capabilities beyond single-qubit demonstrations.

Common Beginner Mistakes

- "Two qubits are just two separate single qubits." In reality, once combined they form a single quantum system described by a larger state vector
- "Two qubits store twice as much information." In reality, they provide a state space with four basis states, not just double the information
- "Tensor product means adding qubits." In reality, tensor products combine quantum systems by multiplying their state spaces

Summary: What to Remember

- Two qubits have four computational basis states: |00>, |01>, |10>, |11>
- Multi-qubit systems are represented using the tensor product, which multiplies state spaces
- The state space grows exponentially as 2^n for n qubits
- The normalisation rule still applies: the total probability across all basis states equals 1
- Multiple qubits are the foundation for controlled gates and entanglement